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Chronic Kidney Disease Renal failure:
Clinical findings, diagnosis and treatment

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Objectives

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1. Understand the causes of progressive loss of renal function and its consequences
2. Classify chronic kidney disease using staging criteria
3. Recognize clinical signs and symptoms of chronic renal failure
4. Understand key diagnostic tests used in staging and evaluating chronic renal failure
5. Identify modalities for renal replacement therapy such as dialysis and transplant

Definition

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Chronic kidney disease (CKD) is the **gradual**, **progressive** **loss** of the kidney's ability to:

- excrete wastes
- concentrate urine
- conserve electrolyte balance

The Kidney Disease: Improving Global Outcomes (KDIGO) CKD Workgroup, 2013, defined CKD as abnormalities of kidney structure or function present for >3 months with implications for health

Brief pathophysiology

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- Each kidney has approximately 1 million nephrons
- There is great capacity for compensation when injured
  - Hyperfiltration and hypertrophy
    - Vasodilatation increases capillary pressure and radius
    - Decreases GFR, lowers kidney function
    - Loss of 50% of functioning nephrons → Plasma creatinine doubles
    - Prolonged hyperfiltration and hypertrophy leads to progressive renal dysfunction
    - Increased glomerular capillary pressure leads to capillary damage which over time leads to progressive glomerular injury: focal → segment → global

Destruction of nephrons leads to compensatory hypertrophy and supranormal GFR of the remaining nephrons in an attempt to maintain overall homeostasis.
As a result, serum creatinine may appear normal even in the face of significant loss of renal mass, making it an insensitive marker for early renal damage and scarring.
In addition, compensatory hyperfiltration leads to overwork injury in the remaining nephrons, which over time leads to glomerular sclerosis and interstitial fibrosis.

Brief pathophysiology

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- Clearance maintained but…

Epidemiology

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In the U.S. based on data from NHANES for 2017 to March 2020:

- 14.0% of adults had low eGFR (eGFR <60 ml/min/1.73m2), albuminuria, or both
  - 10.5% had moderate risk disease, 2.4% high risk disease, and 1.1% very high risk disease.
- CKD prevalence highest in Black individuals (18.8%) and lowest in Hispanic individuals (12.0%)
- Substantially higher percentage of Black individuals than White individuals had a urine Albumin to Creatinine Ration (ACR) ≥30 mg/g: 13.5% versus 9.2%.
- Females were relatively overrepresented for those with CKD - 56.7% of individuals with CKD were female versus 50.9% of individuals without CKD.
- The prevalence of diabetes mellitus (DM) has increased since 2005, reaching 9.5% in those without CKD and 35.6% in those with CKD in 2017-March 2020.
- People with lower levels of income and educational achievement were more likely to have CKD.
- The risk for CKD increases as we age, independent of risk factors like HTN and diabetes
  - It affects all age groups but the percentage of individuals with CKD who were aged ≥65 was 48.7%

Reference:2024 Annual report NIDDK USRDS: https://usrds-adr.niddk.nih.gov/2024/chronic-kidney-disease/1-ckd-in-the-general-population

The 2024 report from the National Institute of Diabetes and Digestive and Kidney Diseases or NIDDK reports on data collected from the National health and nutrition examination survey (NHANES) that was collected from 2017 to March 2020.

This is the most recent data set that has been published.

The percentage of U.S. adults with CKD has been relatively stable since 2005-2008 and was 14.0% in 2017-March 2020. The prevalence of CKD in adults aged ≥65 years decreased steadily, from 38.0% in 2005-2008 to 33.2% in 2017-March 2020 most likely aided by earlier diagnosis and intervention.

A higher percentage of women (15.4% in 2017-March 2020) than men (12.6%) have CKD, a finding that has been consistent since 2005-2008.
In the most recent survey period, CKD prevalence was highest among Black individuals (18.8%) and lowest among Hispanic individuals (12.0%), with White individuals in between (13.8%).

CKD – Risk factors

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Table 5 | Risk factors for CKD

Domains	Example conditions
Common risk factors	Hypertension Diabetes Cardiovascular disease (including heart failure) Prior AKI/AKD
People who live in geographical areas with high prevalence of CKD	Areas with endemic CKDu Areas with the high prevalence of *APOL1* genetic variants Environmental exposures
Genitourinary disorders	Structural urinary tract disease Recurrent kidney calculi
Multisystem diseases/chronic inflammatory conditions	Systemic lupus erythematosus Vasculitis HIV
Iatrogenic (related to drug treatments and procedures)	Drug-induced nephrotoxicity and radiation nephritis
Family history or known genetic variant associated with CKD	Kidney failure, regardless of identified cause Kidney disease recognized to be associated with genetic abnormality (e.g., PKD, *APOL1*-mediated kidney disease, and Alport syndrome)
Gestational conditions	Preterm birth Small gestational size Pre-eclampsia/eclampsia
Occupational exposures that promote CKD risk	Cadmium, lead, and mercury exposure Polycyclic hydrocarbons Pesticides

Reference:2024 KDIGO guidelines: CKD Evaluation and Management – KDIGO

Many things can lead to renal injury that over time lead to chronic kidney injury/disease – however, over 70% of cases of end stage kidney disease (ESKD) in the United States are due to diabetes mellitus or hypertension/vascular disease.

CKD – Etiology, risk

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Summary of causes of CKD	
Vascular disease/disorders	Renal artery stenosis, vasculitides: (+/-) ANCA leading to nephrosclerosis; Atheroembolic causes: renal vein thrombosis, HTN nephrosclerosis
1° glomerular disease	Membranous nephropathy, IgA nephropathy (Berger's), Focal segmental glomerulosclerosis (FSGS), Rapidly progressive GN (RPGN), Minimal change disease, Membranoproliferative GN
2° glomerular disease [most common cause of CKD]	DM, SLE, RA, MCTD, Scleroderma, Goodpasture syndrome, Wegener granulomatosis, Post-infectious GN, HIV, Syphilis, HBV/HCV, IV drug use, Amyloidosis, Henoch-Schönlein purpura
Tubulointerstitial disease	Drugs (i.e, sulfa, allopurinol), infection (viral, bacterial, parasitic), Sjögren syndrome, chronic hypokalemia, chronic hypercalcemia
Urinary tract obstructive disease	Urolithiasis, BPH, tumors (by obstruction), retroperitoneal fibrosis, Urethral stricture, neurogenic bladder

CKD - Diagnosis

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- Chronic kidney disease
  - Signs of kidney disease
  - eGFR < 60 cc/min/1.73m^2 for >3 months
- Kidney Damage
  - Positive markers of damage
    - Abnormal UA
      - proteinuria
    - Abnormal blood tests
    - Abnormal imaging (US/CT/MRI)
    - Abnormal biopsy
      - pathologic changes

Test people at risk for and with chronic kidney disease (CKD) using both urine albumin measurement and assessment of glomerular filtration rate (GFR).

Following incidental detection of elevated urinary albumin-to-creatinine ratio (ACR), hematuria, or low estimated GFR (eGFR), repeat tests to confirm presence of CKD.

Proof of chronicity (duration of a minimum of 3 months) can be established by:

1. review of past measurements/estimations of GFR;
2. review of past measurements of albuminuria or proteinuria and urine microscopic examinations;
3. imaging findings such as reduced kidney size and reduction in cortical thickness;
4. kidney pathological findings such as fibrosis and atrophy;
5. medical history, especially conditions known to cause or contribute to CKD;
6. repeat measurements within and beyond the 3-month point.

Do not assume chronicity based upon a single abnormal level for eGFR and ACR, as the finding could be the result of a recent acute kidney injury (AKI) event or acute kidney disease (AKD).

Interpretation of serum creatinine level requires consideration of dietary intake - high

protein vs lower protein.

For initial testing of albuminuria, a urine albumin to creatinine ratio based off of a collected urine sample is preferred over a reagent strip analysis. In all cases, a first void in the morning midstream sample is preferred in adults and children. Confirm albumin to creatinine ratios ≥30 mg/g on a random untimed urine with a subsequent first morning void in the morning midstream urine sample.

Consider initiation of treatments for CKD at first presentation of decreased GFR or elevated ACR if CKD is deemed likely due to presence of other clinical indicators.

CKD - Diagnosis

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Table 12 | Reported examples of substances that may cause analytical interferences in creatinine assays

Jaffe methods	Enzymatic methods
Acetaminophen[196]	Bilirubin[211]
Aspirin[196]	Lidocaine metabolites[212]
Ascorbic acid[197]	Metamizole[196]
Bacterial contamination[198]	N-acetylcysteine[213]
Bilirubin[199,200]	Proline stabilizers, present in intravenous immunoglobulin preparations[214]
Blood-substitute products[201]	Phenindione[215]
Cephalosporins[202,203]	
Fluorescein[204]	
Glucose[205]	
Hemoglobin F[206]	
Ketones/ketoacids[207]	
Lipids[208]	
Metamizole protein[206-209]	
Pyruvate, including that arising from delayed sample processing[143]	
Streptomycin[210]	

Reference:2024 KDIGO guidelines: CKD Evaluation and Management – KDIGO

Some substances can interfere with creatinine assays.

Table 16 | Factors causing biological variation in urine albumin or urine protein

Factor	Falsely elevated ACR or PCR	False decrease in ACR or PCR
Variability in urine albumin or protein		
Hematuria	Increases albumin and protein in the urine	
Menstruation	Increases albumin and protein in the urine	
Exercise[259]	Increases albumin and protein in the urine	
Infection[260,261]	Symptomatic urinary infection can cause production of protein from the organism	
Nonalbumin proteins		Other proteins may be missed by albumin reagent strips
Variability in urinary creatinine concentration		
Biological sex	Females have lower urinary creatinine excretion, therefore higher ACR and PCR	Males have higher urinary creatinine excretion, therefore lower ACR and PCR
Weight[73,160]	Low urinary creatinine excretion consistent with low weight can cause high ACR or PCR relative to timed excretion	High urinary creatinine excretion consistent with high weight can cause low ACR or PCR relative to timed excretion
Changes in creatinine excretion	Lower urinary creatinine excretion with AKI or low-protein intake	High urinary creatinine excretion with high-protein intake or exercise

Reference:2024 KDIGO guidelines: CKD Evaluation and Management – KDIGO

KDIGO recommends using eGFRcr-cystatin C in clinical situations when eGFRcr is less accurate for example in a patient with reduced muscle mass or who have a muscle mass that exceeds norms and where GFR may affect clinical decision-making.

Understand the factors that can cause biological variations in urine albumin or urine protein levels based on physical state, biological sex, weight, etc.

Table 6 | Guidance for the selection of additional tests for evaluation of cause

Test category	Examples	Comment or key references
Imaging	Ultrasound, intravenous urography, CT kidneys ureters bladder, nuclear medicine studies, MRI	Assess kidney structure (i.e., kidney shape, size, symmetry, and evidence of obstruction) for cystic disease and reflux disease. Evolving role of additional technologies (e.g., 3D ultrasound)
Kidney biopsy	Ultrasound-guided percutaneous	Usually examined by light microscopy, immunofluorescence, and electron microscopy, and, in some situations, may include molecular diagnostics Used for exact diagnosis, planning treatment, assessing activity and chronicity of disease, and likelihood of treatment response; may also be used to assess genetic disease
Laboratory tests: serologic, urine tests	Chemistry including acid base and electrolytes, serologic tests such as anti-PLA2R, ANCA, anti-GBM antibodies Serum-free light chains, serum, and urine protein electrophoresis/immunofixation Urinalysis and urine sediment examination	Refer to *KDIGO 2021 Clinical Practice Guideline for the Management of Glomerular Diseases*[22] Increasing recognition of the role of light chains in kidney disease even in the absence of multiple myeloma (monoclonal gammopathy of renal significance [MGRS])[98] Presence of persistent hematuria or albuminuria is critical in determining differential diagnosis
Genetic testing	*APOL1, COL4A3, COL4A4, COL4A5, NPHS1, UMOD, HNF1B, PKD1, PKD2*	Evolving as a tool for diagnosis, increased utilization is expected. Recognition that genetic causes are more common and may present without classic family history[99,100]

ANCA, antineutrophil cytoplasmic antibody; *APOL1*, apolipoprotein 1; *COL4A*, type IV collagen alpha chain; CT, computed tomography; GBM, glomerular basement membrane; *HNF1B*, hepatocyte nuclear factor 1B; MRI, magnetic resonance imaging; *NPHS1*, congenital nephrotic syndrome; *PKD1*, polycystic kidney disease-1; *PKD2*, polycystic kidney disease-2; PLA2R, M-type phospholipase A2 receptor; *UMOD*, uromodulin.

Additional tests to consider when trying to determine the cause of the CKD.

KDIGO staging & prognosis

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CKD is classified based on: • Cause (C) • GFR (G) • Albuminuria (A)				Albuminuria categories Description and range		
				A1	A2	A3
				Normal to mildly increased	Moderately increased	Severely increased
				<30 mg/g <3 mg/mmol	30–299 mg/g 3–29 mg/mmol	≥300 mg/g ≥30 mg/mmol
GFR categories (ml/min/1.73 m²) Description and range	G1	Normal or high	≥90	Screen 1	Treat 1	Treat 3
	G2	Mildly decreased	60–89	Screen 1	Treat 1	Treat 3
	G3a	Mildly to moderately decreased	45–59	Treat 1	Treat 2	Treat 3
	G3b	Moderately to severely decreased	30–44	Treat 2	Treat 3	Treat 3
	G4	Severely decreased	15–29	Treat* 3	Treat* 3	Treat 4+
	G5	Kidney failure	<15	Treat 4+	Treat 4+	Treat 4+

Low risk (if no other markers of kidney disease, no CKD)
Moderately increased risk
High risk
Very high risk

- * Stage 1 or 2 need evidence of kidney damage like
  - Persistent micro albuminuria/proteinuria
  - Persistent hematuria(not otherwise caused)
  - Structural abnormality of kidneys

Reference:2024 KDIGO guidelines: CKD Evaluation and Management – KDIGO

CKD Heat map: https://www.kidney.org/sites/default/files/heat_map_card.pdf

Kidney Disease: Improving Global Outcomes or “KDIGO” is a global organization that develops and implements evidence-based clinical practice guidelines to improve patient care for people with kidney disease worldwide. They developed the tools to assess and classify patient risk for kidney disease progression and complications.

Based on their guidelines, staging should use GFR in conjunction with albumin to creatinine ratio (ACR) to determine risk and prognosis. This is illustrated by the chronic kidney disease heat map”

Reading top to bottom you have the use of GFR to determine the stage (i.e the severity) of kidney disease
Reading from left to right you have the persistent presence of albuminuria as in indicator of renal disease
The colors in the boxes represent the prognosis of chronic kidney disease using the two categories – the highest risk/poorest prognosis is in the lower right hand corner – stage 5 with severely increased albumin to creatinine ratio.

Table 4 | Use of GFR and albuminuria

Clinical decisions	Current level: GFR	Current level: Albuminuria	Change in the level of GFR
Diagnosis and staging	• Detection of CKD • Evaluation for kidney donation	• Detection of CKD	• Detection of AKI and AKD • Detection of CKD progression
Treatment	• Referral to nephrologists • Patient and family education about CKD and benefit of lifestyle changes • Monitor progression of GFR decline • Referral for kidney transplantation • Placement of dialysis access • Dosage and monitoring for medications cleared by the kidney • Determine safety of diagnostic tests or procedures • Eligibility for clinical trials	• Referral to nephrologists • Patient and family education about CKD and benefit of lifestyle changes • Monitor progression of GFR decline • Eligibility for clinical trials	• Treatment of AKI • Monitoring drug toxicity • Re-evaluate CKD treatment strategies
Risk assessment	• Risk of CKD complications • Risk for CKD progression • Risk of CVD • Risk for medication errors • Risk for perioperative complications • Risk for mortality • Fertility and risk of complications of pregnancy	• Risk for CKD progression • Risk for CVD • Risk for mortality • Fertility and risk of complications of pregnancy	• Risk for kidney failure • Risk for CVD, HF, and mortality • Risk for adverse pregnancy outcome

Reference:2024 KDIGO guidelines: CKD Evaluation and Management – KDIGO

Assess albuminuria in adults, or albuminuria/proteinuria in children, and GFR at least annually in people with CKD.
Assess albuminuria and GFR more often for individuals at higher risk of CKD progression when measurement will impact therapeutic decisions.
For people with CKD, a change in eGFR of >20% on a subsequent test exceeds the expected variability and warrants evaluation.
For albuminuria monitoring of people with CKD, a doubling of the ACR on a subsequent test exceeds laboratory variability and warrants evaluation.

Use disease-specific, externally validated prediction equations in people with immunoglobulin A nephropathy (IgAN) and autosomal dominant polycystic kidney disease (ADPKD).

For cardiovascular risk prediction to guide preventive therapies in people with CKD, use externally validated models that are either developed within CKD populations or that incorporate eGFR and albuminuria.

Symptoms related to each stage

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- **Patients are generally asymptomatic until the GFR <30 cc/min**
- Stages 1-3: therefore asymptomatic
  - Except for possible underlying causes
  - Diagnosis by lab results
- Stage 4: may have electrolyte abnormality or early uremia
- Stage 5: established kidney failure with signs of uremia
  - Requires permanent renal replacement therapy (RRT)

Presenting signs/symptoms

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Symptoms of CKD result directly from disruption of kidney function:

- Hypertension
- Uremia
- Urine output
- Flank pain
- Edema

Signs of CKD:

- Look for signs of underlying disease also
- Elevated BP
- Skin dry/signs of scratching
- Mucous membranes dry
- Joints swollen or painful
- Dyspnea
- Drowsiness/looks tired
- Edema

Symptoms of uremia

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- Nausea
- Vomiting
- Diarrhea
- Anorexia
- Myalgia
- Restless Leg
- Fatigue
- Somnolence
- Excessive bruising
- Shortness of breath
- Confusion
- Peripheral Numbness
- Dry skin
- Pruritus
- Erectile dysfunction
- Amenorrhea

Findings on urinalysis

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- UA or urine dipstick
  - + protein
- Spot urine for protein:creatinine ratio
  - Acceptable in place of 24 hour collections
  - >3 = nephrotic disease
  - <2 = tubulointerstitial disease
- Urine for sediments
  - Evaluate for casts and cells

Source: National Kidney Foundation. K/DOQI clinical practice guidelines for chronic kidney disease: evaluation, classification, and stratification. Am J Kidney Dis 2002;39(2 suppl 1):S216

Lab findings in CKD

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- Electrolytes
  - Hyperkalemia
  - Hyperphosphatemia
  - Hypocalcemia
  - Low bicarbonate
  - Hyponatremia (if free water is in excess)
- Elevated BUN and creatinine
- Vitamin D deficiency
- Anemia of chronic disease

Source: National Kidney Foundation. K/DOQI clinical practice guidelines for chronic kidney disease: evaluation, classification, and stratification. Am J Kidney Dis 2002;39(2 suppl 1):S216

As CKD increases in severity, electrolyte abnormalities and hormonal changes occur. More notable changes include hypocalcemia, hyperphosphatemia, hyperkalemia with metabolic acidosis, hyperparathyroidism, decreased erythropoietin, and reduced levels of activated vitamin D.

Anemia of chronic disease

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- Work-up for anemia when
  - Hgb < 11 in premenopausal females
  - Hgb < 12 in adult males and post-menopause
- Initial work-up should evaluate for
  - Iron deficiency
  - RBC indices
  - Reticulocyte counts
- Erythropoietin level not indicated
- Normochromic, normocytic anemia
  - Low erythropoietin level
  - Decreased hemoglobin/hematocrit
  - TIBC is low, Fe is low, ferritin can be high
  - Normal RDW sizing, MCV is normal
  - Reticulocyte count will be normal
- Can be microcytic anemia if prolonged
  - MCV is decreased

Electrolyte disturbances

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Hyperkalemia

- Potassium excretion: dependant on distal flow and aldosterone secretion
- Oliguric pts, hypoaldosteronism, increased tissue breakdown, high K diet
- Reduce dietary intake and avoid NSAIDs

Metabolic acidosis

- Tendency to retain hydrogen ions
  - Bicarb stabilizes between 12-20 meq/L
- Treatment is with alkali (i.e. sodium bicarbonate) supplement orally
  - Maintain bicarbonate >23meq/L
  - May require dialysis

Electrolyte disturbance

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Hyperphosphatemia

- Phosphate retention

Causes secondary hyperparathyroidism

- PTH initially corrects and maintains GFR >30
- Prolonged high PTH causes renal osteodystrophy

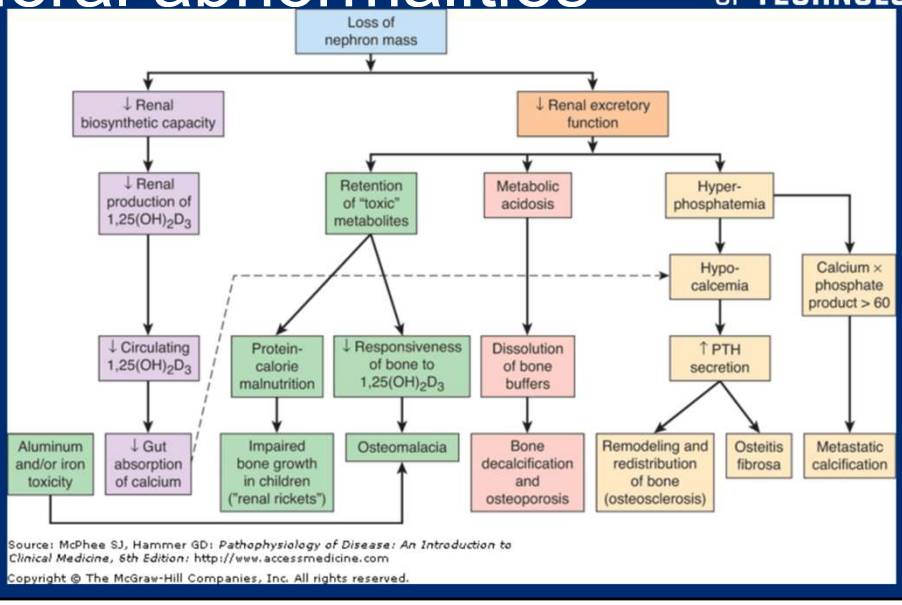
Restriction of phosphate in diet (800 mg/day)

- Does not decrease serum phosphate levels
- DOES improve secondary hyperparathyroidism

Phosphate binders may be used

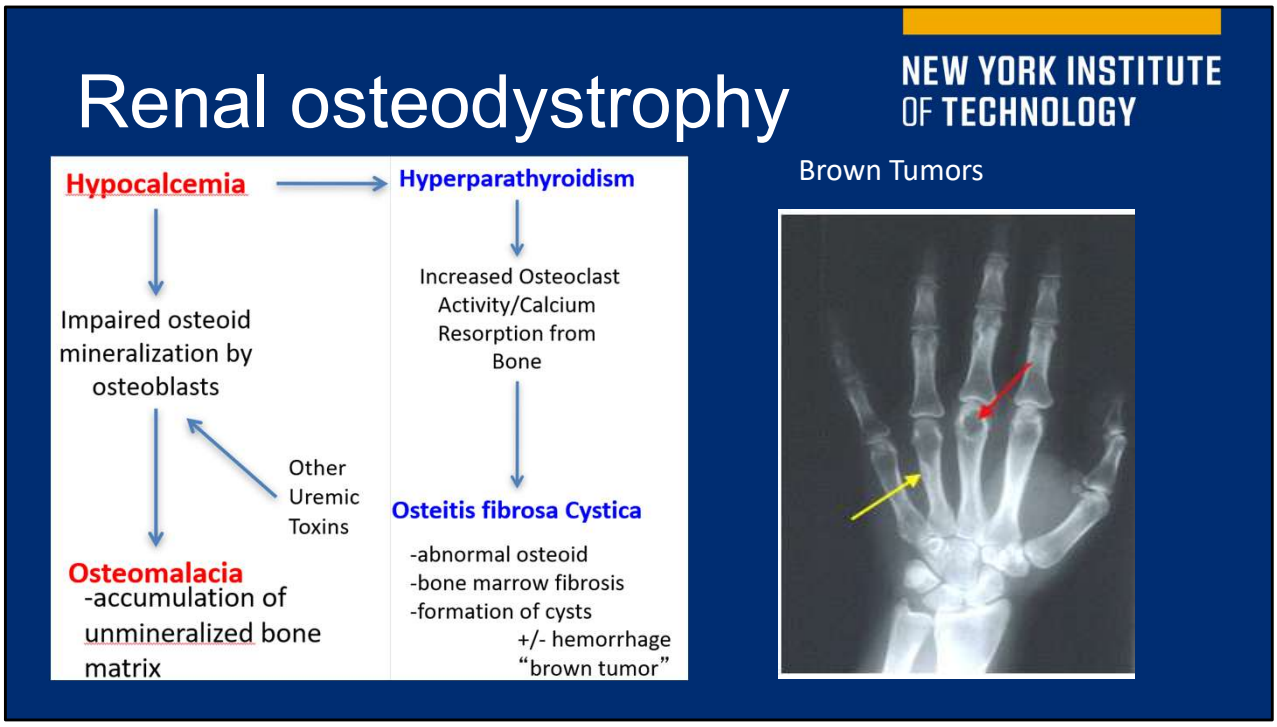
Mineral abnormalities

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Source: McPhee SJ, Hammer GD: *Pathophysiology of Disease: An Introduction to Clinical Medicine, 6th Edition*: http://www.accessmedicine.com

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Renal osteodystrophy is caused by secondary hyperparathyroidism, which can be distinguished from primary hyperparathyroidism by the presence of hypocalcemia instead of hypercalcemia. The hypocalcemia is due to (1) hyperphosphatemia with consequent complexing of calcium in the serum, and (2) the decreased conversion of vitamin D by the kidneys, which is needed to increase serum calcium levels. The chronic hypocalcemia leads to secondary hyperparathyroidism. In this setting, hypocalcemia is associated with tetany, seizures, QT prolongation, twitching (Chvostek sign), and spasm (Trousseau sign). An example of Chvostek sign is the initiation of muscle twitching when the facial muscles are tapped. An example of Trousseau sign is the presence of muscle spasms when the arm is squeezed. The seizures associated with hypocalcemia can lead to injury including fractures while the QT prolongation in patients with hypocalcemia due to chronic kidney disease (CKD) can lead to life-threatening arrhythmias such as torsade de pointes and significantly increases the risk of sudden cardiac death

Renal imaging

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- **X-ray (plain film)**
  - Find radio-opaque stones
  - Nephrocalcinosis
- **Voiding cystourethrogram**
  - Evaluate for ability/disabilty to void
  - Seen in neurogenic bladder
- **Renal ultrasound**
  - Evaluate renal size (may be normal in most disease)
  - Hydronephrosis
  - Polycystic kidney disease
  - May find compressive tumors
- **Nuclear flow imaging**
  - Evaluate for Renal artery stenosis

Renal Imaging

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- **CT**
  - Evaluates for structural disease
  - May differentiate findings from U/S
  - Dye may make it contraindicated in stages 3-5
  - Caution in patients with diabetes on certain meds
- **MRI**
  - Option instead of CT to avoid problems with dye
  - Able to evaluate for thrombosis
- **Angiography**
  - Standard for diagnosis of renal artery stenosis

Source: National Kidney Foundation: https://www.kidney.org/kidney-topics/contrast-dye-and-kidneys

CT scans with iodinated contrast – this type of contrast dye can cause acute kidney injury (AKI). Another name for this is "contrast-associated acute kidney injury" or CA-AKI. The risk for most people is low, but it can be higher if the pt has advanced CKD, diabetes, dehydration, etc.

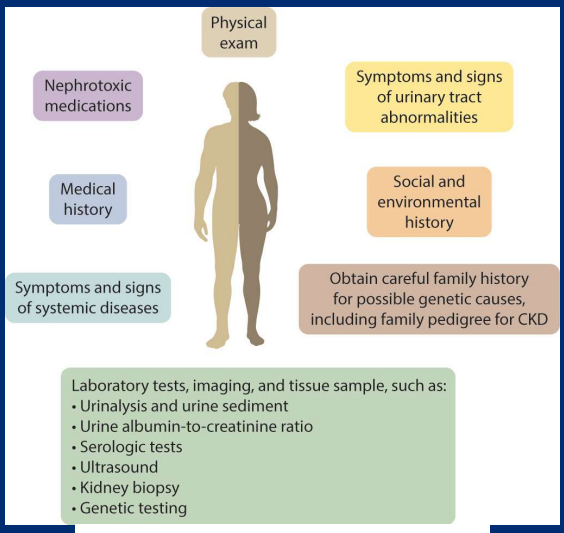
Some of the medications to stop if a patient with CKD needs to get a CT with contrast:
Metformin: Due to the risk of lactic acidosis in patients with decreased kidney function, metformin is typically stopped the day before the CT if possible and it is often restarted after the test, provided kidney function has not worsened.
NSAIDs: should be stopped 24 to 48 hours before a contrast study.
Diuretics: these include loop and thiazide-like diuretics -should be stopped at least 24 hours before a contrast study, if possible.
Renin-angiotensin system (RAS) blockers: includes ACE inhibitors and ARBs. While there is ongoing debate about the evidence for stopping these, clinical guidelines often recommend it to reduce the risk of contrast-induced nephropathy (CIN)

Primary prevention of CKD

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Primary Prevention

- Identify patients that could be higher risk
- Encourage healthy lifestyles
  - Diet, exercise, weight loss
  - Avoidance of substance abuse



Reference:2024 KDIGO guidelines: CKD Evaluation and Management – KDIGO

Primary prevention aims to prevent disease or injury before it ever occurs. This is done by preventing exposures to hazards that cause disease or injury, altering unhealthy or unsafe behaviours that can lead to disease or injury, and increasing resistance to disease or injury should exposure occur.

Identify patients that are at higher risk for CKD and screen using GFR and albumin/creatining ration (ACR):
Beyond hypertension and diabetes, risk factors include being overweight/obese, age (being over 60 years old), and having a family history of kidney disease.
Hypertension and diabetes are the top two risk factors for CKD, and yet a strong majority of people with these conditions have not had the tests needed to diagnose CKD, including: 94% of people with hypertension, 61% of people with diabetes, and 60% of people with BOTH hypertension and diabetes. According to the National Kidney Foundation, up to 90% of patients are unaware that they have CKD and even if the testing is done, it does not always lead to a referral to a nephrologist or discussion about their risks and how to reduce their risk.

National Kidney Foundation has a program called “CKD Intercept” which aims to improve primary prevention by helping communities and primary care providers improve screening for CKD, improve education and management, and reduce advancement of kidney disease.

https://www.kidney.org/professionals/ckdintercept

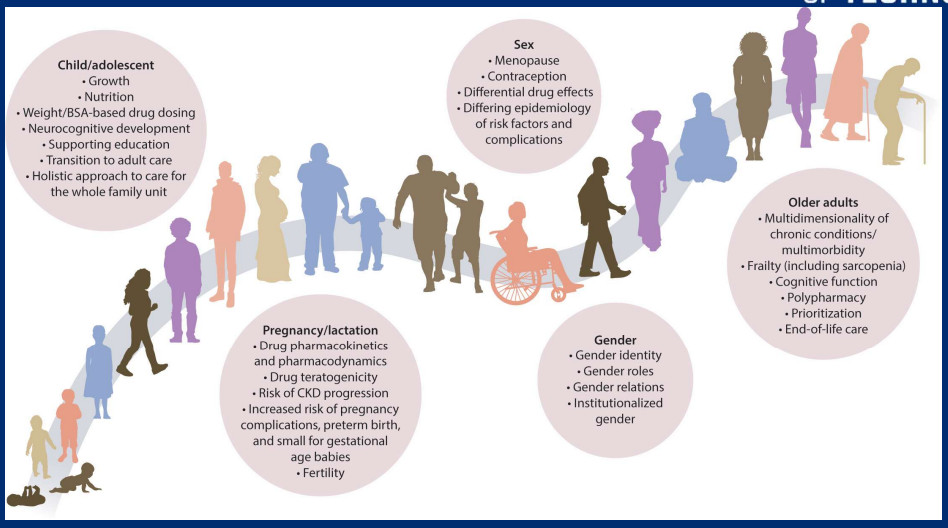
EVALUATION

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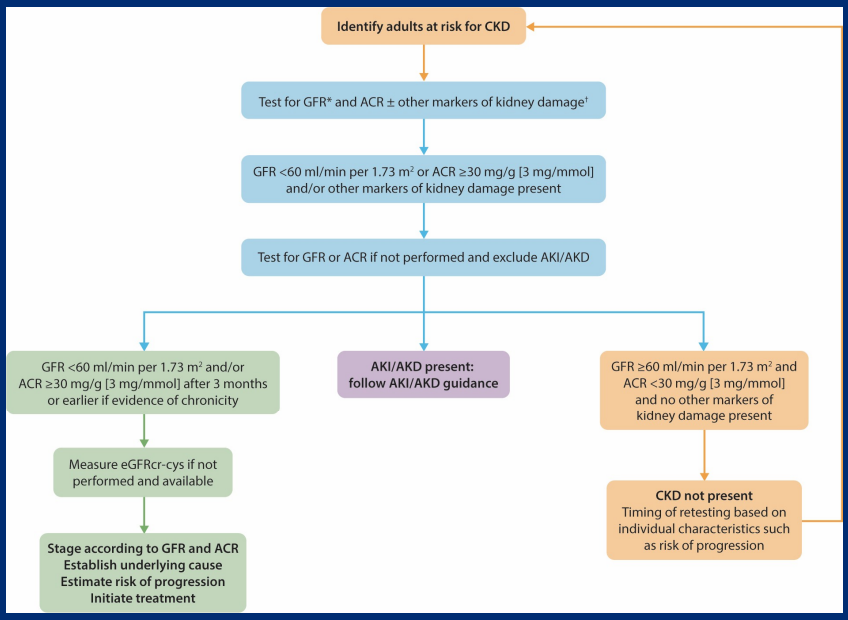
Use a personalized approach to diagnosis, risk assessment, and management that considers age, sex, and gender.

At the extremes of age - the very young and the very old – goals, treatment modalities, and decision-making differ due to differences in prognosis, treatment options, and prioritization.

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Reference:2024 KDIGO guidelines: CKD Evaluation and Management – KDIGO



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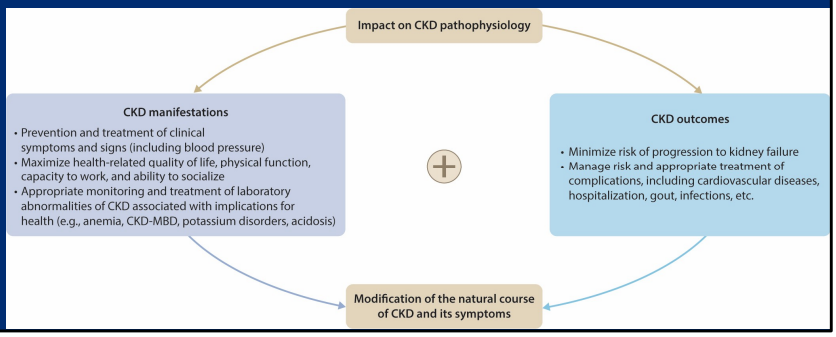
Reference:2024 KDIGO guidelines: CKD Evaluation and Management – KDIGO

Management

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Goals:

- Delay progression to end stage renal kidney disease (EKSD)(stage 5)
- Treat associated symptoms of renal failure
- Avoid damage to other organ systems
- Limit disability and effect on quality of life



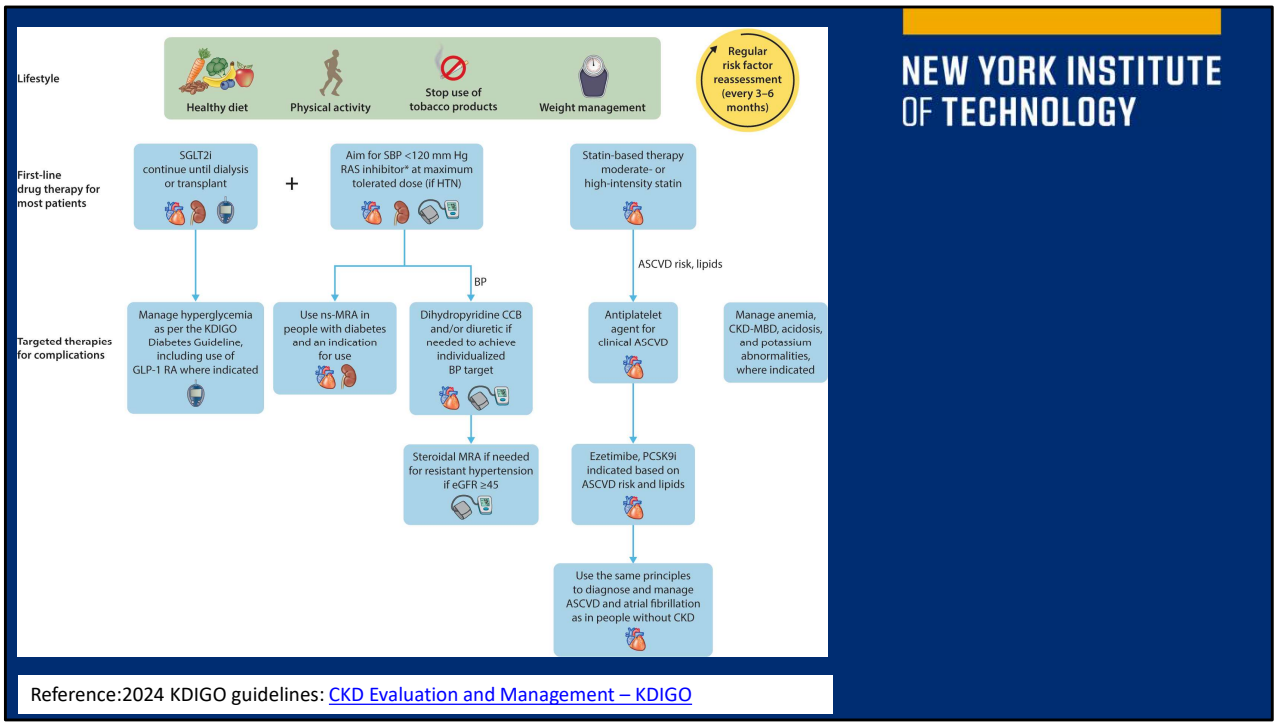
Reference:2024 KDIGO guidelines: CKD Evaluation and Management – KDIGO

Treat people with CKD with a comprehensive treatment strategy to reduce risks of progression of CKD and its associated complication

Encourage people with CKD to

1. undertake physical activity compatible with cardiovascular health, tolerance, and level of frailty; KDIGO recommends that people with CKD be advised to undertake moderate intensity physical activity for a cumulative duration of at least 150 minutes per week, or to a level compatible with their cardiovascular and physical tolerance. Recommendations for physical activity should consider age, background, presence of other comorbidities, and access to resources. People with CKD should be advised to avoid sedentary behavior. For people at higher risk of falls, healthcare providers should provide advice on the intensity of physical activity (low, moderate, or vigorous) and the type of exercises (aerobic vs. resistance, or both).
2. Aim to lose weight to achieve an optimal body mass index (BMI) through healthy eating and exercise;
3. Stop the use of tobacco products.

Referral to providers and programs (e.g., psychologists, renal dietitians or accredited nutrition providers, pharmacists, physical and occupational therapy, and smoking cessation programs) should be offered where indicated and available.



Adopting a healthy and diverse diet with a higher consumption of plant-based foods compared to animal-based foods and a lower consumption of ultra-processed foods has the potential to benefit complications related to progressive CKD such as acidosis, hyperkalemia, and hyperphosphatemia with less risk of protein energy-wasting.

KDIGO recommends maintaining a protein intake of 0.8 g/kg body weight per day in adults with CKD stage 3 to stage 5. Avoid high protein intake (>1.3 g/kg body weight per day) in adults with CKD at risk of progression. For those who are at risk of kidney failure, providers can prescribe a lower protein diet. Do not prescribe low- or very low–protein diets in metabolically unstable people with CKD.

Do not restrict protein intake in children with CKD due to the risk of growth impairment.

KDIGO recommends that sodium intake be <2 g of sodium per day in adults with CKD and that adults with HTN and CKD be treated to a target systolic blood pressure (SBP) of <120 mm Hg,
when tolerated. They also recommend considering less intensive BP-lowering therapy in adults with frailty, high risk of falls and fractures, very limited life expectancy, or symptomatic postural hypotension.

ARBs, ACE inhibitors, SGLT-2 inhibitors, and mineralocorticoid receptor antagonists (MRA) help reduce hyperfiltration, which can slow progression of CKD.
Treatments that delay progression of CKD with a strong evidence base include Renin-Angiotensin System Inhibitors (RASi) and SGLT2 inhibitors. In people with CKD and heart failure, SGLT2 inhibitors confer benefits irrespective of levels of albuminuria. (Benefits of SGLT2 inhibitors: lower blood sugar levels, reduce the risk of cardiovascular events in people with type 2 diabetes, promote weight loss, and protect kidney function. Common SGLT2 inhibitor medications: Empagliflozin (Jardiance), Dapagliflozin (Farxiga), and Canagliflozin (Invokana).)

Management

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Secondary Prevention

- Initiate aggressive treatment
  - Targeted drug regimens known to prevent progression to renal disease
    - **E.g., ACE-Inhibitors in DM**
- Screening for renal involvement including imaging
- Continued medication management
- Control disease states that can contribute

Resources for patients: https://www.kidney.org/kidney-topics/chronic-kidney-disease-ckd

Secondary prevention aims to reduce the impact of a disease or injury that has already occurred. This is done by detecting and treating disease or injury as soon as possible to halt or slow its progress, encouraging personal strategies to prevent reinjury or recurrence, and implementing programs to regain some if not all of their health and function to prevent long-term problems.

The goal is to prevent further advancement of renal disease and treat/manage what is already present that can lead to worsening renal function such as control hypertension, improve diabetic control, and manage lifestyle as much as possible – lose weight, improve diet, exercise, stop smoking/alcohol, etc. Reinforce goals and education can help patients: use resources available online – for example from the National Kidney Foundation.

For example:
In Diabetic Patients: we want to have aggressive glycemic control with HbA1c consistently less than 7%; Role of ACE-Inhibitors for diabetics: there is demonstrated benefit of using ACE-I when proteinuria is found but not necessarily in the earliest stages of kidney disease; have to monitor potassium levels and will have to avoid ACE-I in advanced CRF (stage 4-5) as well as patient with renal artery stenosis
Initiate medications that can help prevent further renal injury: KDIGO recommends treatment with a statin or statin/ezetimibe combination in adults aged ≥50 years with eGFR

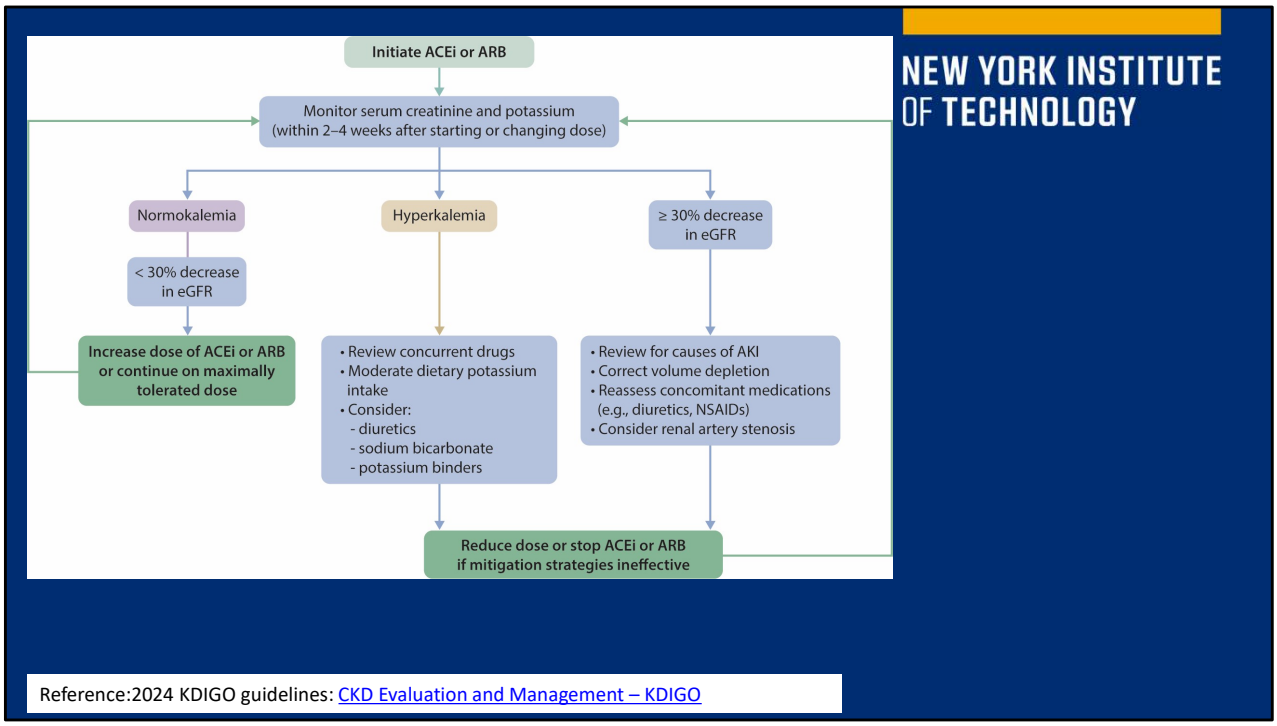
<60 but not treated with chronic dialysis or kidney transplantation. In adults aged ≥50 years with CKD and eGFR ≥60, they recommend treatment with a statin. In adults aged 18–49 years with CKD but not treated with chronic dialysis or kidney transplantation, they recommend statin treatment in people with one or more of the following:

- known coronary disease (myocardial infarction or coronary revascularization),
- diabetes mellitus,
- prior ischemic stroke, or
- estimated 10-year incidence of coronary death or nonfatal myocardial infarction >10%.

KDIGO also recommended considering oral low-dose aspirin for prevention of recurrent ischemic cardiovascular disease events (i.e., secondary prevention) in people with CKD and established ischemic cardiovascular disease; if cannot tolerate aspiring then consider use of other antiplatelet therapy (e.g., P2Y12 inhibitors).
ARBs, ACE inhibitors, SGLT-2 inhibitors, and mineralocorticoid receptor antagonists (MRA) help reduce hyperfiltration, which can slow progression of CKD.

Avoid medications/ treatments that may further damage kidney function; Perform thorough medication review periodically and at transitions of care to assess adherence, continued indication, and potential drug interactions because people with CKD often have complex medication regimens and are seen by multiple specialists. Review and limit the use of over-the-counter medicines like NSAIDs, dietary, or herbal remedies that may be harmful for people with CKD. Monitor eGFR, electrolytes, and therapeutic medication levels, when indicated, in people with CKD receiving medications with narrow therapeutic windows, potential adverse effects, or nephrotoxicity, both in outpatient practice and in hospital settings.

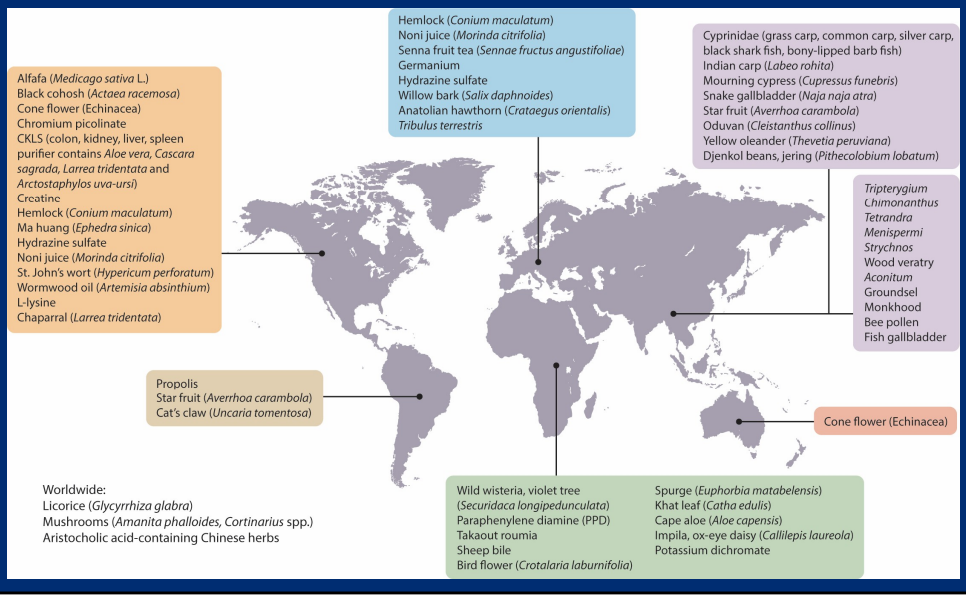
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Once initiated on ACEi or ARB, the patient must be monitored carefully including assessing eGFR and electrolytes after initiation of medication and on a regular basis – for example, assess creatinine and potassium levels within 2 to 4 weeks of starting a dose or changing a dose. If a patient develops, hyperkalemia – the patient will require investigation and efforts at mitigation to prevent continued or worsening hyperkalemia and may necessitate stopping the ACEi or ARB if mitigation strategies are not effective.

Potentially nephrotoxic dietary and herbal remedies

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Reference:2024 KDIGO guidelines: CKD Evaluation and Management – KDIGO

Management

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Tertiary prevention:

- Staging of chronic renal impairment
- Treat disease and/or associated symptoms
- Plan for future - such as need for dialysis, renal transplant

Tertiary prevention aims to soften the impact of ongoing illness that has lasting effects and that if not managed can lead to significant loss of quality of life and early disability or death. This is done by helping people manage long-term, complex health problems in order to improve as much as possible their ability to function, their quality of life and their life expectancy.

In this case, the goal would be to manage the symptoms/signs of advanced chronic kidney disease, continue to control other chronic conditions, begin to prepare for the need for more advanced treatments (consult with nephrology, nutritionists/dietiticians, vascular surgeon if have not yet done so, and consider having pt meet with renal transplant team) and/or implement advanced treatments such as dialysis.

Plans addressing future health care states should be jointly agreed with the patient and their families/caregivers. Advanced care planning for those choosing advanced care is particularly important and should involve kidney transplant teams and nephrologists as part of a multidisciplinary approach.

Ask people with progressive CKD about uremic symptoms (e.g., reduced appetite, nausea, and level of fatigue/lethargy) at each consultation using a standardized validated assessment of uremic symptoms tool.

Screen people with CKD stage 4 or 5, aged >65, poor growth (pediatrics), or symptoms such as involuntary weight loss, frailty, or poor appetite twice annually for malnutrition using a validated assessment tool.

Enable availability of appropriate medical nutrition therapy for people with signs of malnutrition, ideally under the supervision of renal dietitians or accredited nutrition providers if not available.

Management: Renal replacement therapy

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- Renal Replacement Therapy
  - Preparation for kidney replacement
    - Stage 4 - GFR 15-29 mL/min
  - Kidney replacement treatments
    - Stage 5 – GFR <15 mL/min
- Patients should be educated about disease progression and anticipatory care throughout their care

- Dialysis
  - Hemodialysis
    - Vascular access should be placed at least 6 months prior to anticipated need
    - Takes ~ 3-4 months for AV fistula to mature
  - Peritoneal Dialysis
- Renal transplantation
  - Best potential for complete recovery
  - Preemptive transplant
  - Early transplant
  - Most will end up on dialysis before a transplant

Dialysis removes extra fluid and waste products from the blood when the kidneys are not able to do so properly. Dialysis has been a standard treatment for kidney failures since the 1970s. With end stage kidney disease (when 10-15% of kidney function remains, measured by an estimated glomerular filtration rate (eGFR) of less than 15 mL/min), the kidneys are no longer able to remove waste and fluids properly, retain/excrete electrolytes or minerals properly, or help with regulation of blood pressure without assistance. Dialysis is able to do some of the work of the kidneys, but it is not a cure for kidney disease. With end stage kidney disease, the patient will need dialysis for the rest of their life or until they are able to get a kidney transplant.

In hemodialysis, a dialyzer is used to remove waste and extra fluid from the blood, and then return the filtered blood back to the body. Before starting hemodialysis, a minor surgery is needed to create a vascular access site usually in the patient's arm. Dialysis can be done in a hospital, a dialysis center, or at home. Treatments usually last about four hours and are done three times per week. Side effects include fatigue, hypotension, muscle cramps, weakness, nausea, blood loss, risk for infection at the vascular site (fistula), and pruritus.

In peritoneal dialysis, the blood is filtered inside the body instead of using a dialyzer machine. For this type of dialysis, the lining of the abdomen (the peritoneum) is used as a

filter. Before starting peritoneal dialysis, a minor surgery is needed to place a catheter into the patient's abdomen. During each treatment, the abdomen is slowly filled with dialysate through the catheter. As the patient's blood flows naturally through the area, extra fluid and waste products are pulled out of the blood vessels and into the abdominal area by the dialysate. After a few hours, the fluid mixture is drained from the abdomen using the same catheter and bag that was used at the beginning of the treatment. Peritoneal dialysis can be done almost anywhere so long as the patient has the supplies required to perform the treatment. Side effects include hernia at the catheter access site, weight gain, infection at the catheter site, fatigue and pruritus.

The need for a kidney transplant is the inevitable outcome of chronic kidney disease. There are two types of kidney transplants: a deceased donor kidney transplant and a living donor kidney transplant. A deceased donor kidney comes from someone has died and made the decision to become an organ donor. A kidney from a living donor is from someone who is living, has two healthy kidneys, and chooses to donate one to someone in need. Depending on age, many people will need more than one kidney transplant during a lifetime.

Preemptive transplant: getting a transplant before the patient needs to start dialysis.
Early Transplant: getting a transplant not long after the kidneys fail but with shorter time on dialysis.

People who get a preemptive or early transplant receive their kidney when their health is generally good, which allows them to stay healthier and live longer despite needing a transplant.
According to studies, getting a preemptive or early transplant has several benefits, including:

- Save time and money
- Decreased risk of rejection
- Most people live longer and have a better quality of life
- Avoid being on dialysis for prolonged periods
- Have a less restrictive diet than they would have on dialysis
- Most people stay healthier and avoid some of the health problems associated with dialysis

Post transplant

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- Malignancy
  - 5-6% of patient on immunosuppressive therapy will develop cancer
- Hypercalcemia
- Hypertension
- Hepatitis
  - Immunosuppressive therapy decreases immune control of Hepatitis B and C (test prior to transplant to counsel)

Patients who have undergone renal transplant will have to be on immunosuppressive therapy and should be educated about potential complications and/or risks that could develop after transplant. These patients are managed by a team of specialists and closely monitored.

Morbidity/mortality

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- Hospitalization and increased morbidity
  - Dialysis patients average 2 hospital admissions/yr
  - After renal transplant average 1 admission/yr
- 5 year survival ESRD = 35%
  - Coupled with DM = 25%
- Transplantation increases the survival of patients with stage 5 CKD significantly
- Most common cause of death in CKD patients is Cardiovascular Disease

Outcomes after renal transplant

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- Improved quality of life and decreased disability when compared to dialysis
- Improved life expectancy
  - 5 year survival rate of 34% in dialysis population
- Highest mortality post transplant is in the 1st year

	1 year follow up	5 year follow up	10 year follow up
Deceased donor	95%	81%	61%
Living donor	98%	90%	76%

Case 1

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A 47 year-old male is diagnosed with hypertension at his annual physical exam. On his CMP, the eGFR is 88 so his PCP orders an albumin/creatinine ratio which turns out to be less than 30.

How would we stage his level of CKD?

What guidance would we give him at this time?

Case 2

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A 57 year-old male was diagnosed with diabetes type 2 and Stage 2 CKD when he was 50 years old. He was not compliant with lifestyle recommendations, did not control his diabetes adequately and did not return for follow-up until recently because he was feeling tired all the time and had numbness in his feet. At this visit lab work was drawn, and it was discovered that the eGFR is 25 and his albumin/creatinine ratio is over 300.

How would we stage his level of CKD?

What guidance would we give him at this time?

Summary

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Chronic Kidney Disease is the gradual, progressive loss of the kidney's ability to function properly. It is important to recognize and identify those who are at increased risk, screen for CKD and stage it properly so that we can mitigate the risk of developing chronic kidney disease, educate our patients on lifestyle and management options, and get patient's the treatment needed to manage CKD and slow its advance as much as possible. In instances where CKD has progressed to end stage kidney disease, patients may need renal replacement therapy either through dialysis or via a renal transplant.

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Office hours most Mondays 12 pm – send an email to confirm hours on a specific day

Evaluations:
https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R

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